ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Quadratic Equations** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $\frac{9}{2}$	Sum $= -b/a$
2	**(c)** 2	Product $= c/a$
3	**(a)** 7	Roots are 2, 9
4	**(b)** 20	$16 - 36 + k = 0$
5	**(b)** two distinct real roots	$D = 4$
6	**(b)** 8	Factorise
7	**(a)** $-\frac{7}{3}$	Sum $= -b/a$
8	**(d)** -2	Product $= c/a$
9	**(b)** 8	Roots are -3, 5
10	**(c)** 30	$25 - 55 + k = 0$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$D = (-9)^2 - 4(2)(4) = 49$ (1); $x = \frac{9 \pm 7}{4}$ (1); roots $\frac{1}{2}, 4$ (1).	1+1+1
12	Put $x = 4$: $16 - 36 + k = 0$ (1); $k = 20$ (1); other root $= 9 - 4 = 5$ (1).	1+1+1
13	$D = 0$: $(6)^2 - 4k(1) = 0$ (1); $k = 9$ (1).	1+1
14	$D = (7)^2 - 4(3)(-6) = 121$ (1); $x = \frac{-7 \pm 11}{6}$ (1); roots $\frac{2}{3}, -3$ (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Multiply by $x(x+1)$: $2x + 1 = \frac{5}{6}x(x+1)$ (1); form the quadratic with integer coefficients (1); positive root $x = 2$ (1).	1+1+1
16	$D = 25$ (1); two distinct real roots (1); roots $2, \frac{1}{3}$ (1).	1+1+1
17	$D = (-10)^2 - 4(1)(21) = 16$ (1); $x = \frac{10 \pm 4}{2}$ (1); roots $x = 7, 3$ (1).	1+1+1
	OR: Multiply by $x(x+1)$: $2x + 1 = \frac{7}{12}x(x+1)$ (1); form the quadratic with integer coefficients (1); positive root $x = 3$ (1).	1+1+1
18	$D = -7$ (1); no real roots (1); no real roots exist (1).	1+1+1
	OR: $D = (2)^2 - 4(1)(-8) = 36$ (1); $x = \frac{-2 \pm 6}{2}$ (1); roots $x = 2, -4$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Speed x: $\frac{200}{x} - \frac{200}{x+5} = 2$ (1); $1000 = 2x(x+5)$ so $x^2 + 5x - 500 = 0$ (2); $(x-20)(x+25) = 0$ (1); speed 20 km/h (1).	1+2+1+1
	OR: Base x, altitude $x - 1$: $x^2 + (x-1)^2 = 841$ (1); $2x^2 - 2x - 840 = 0$, i.e. $x^2 - x - 420 = 0$ (2); $(x-21)(x+20) = 0$ (1); base 21 cm, altitude 20 cm (1).	1+2+1+1
20	Speed x: $\frac{280}{x} - \frac{280}{x+5} = 1$ (1); $1400 = x(x+5)$ so $x^2 + 5x - 1400 = 0$ (2); $(x-35)(x+40) = 0$ (1); speed 35 km/h (1).	1+2+1+1
	OR: Boat speed x: $\frac{15}{x-2} + \frac{15}{x+2} = 4$ (1); $15(2x) = 4(x^2 - 4)$ (1); roots of the resulting quadratic: $x = 8$ or negative (1); speed 8 km/h (1).	1+1+1+1

Mistakes to avoid:

- Writing the formula as $\frac{-b \pm \sqrt{b^2 + 4ac}}{2a}$ or dividing only part of the numerator by $2a$.
- Forgetting the sign of b: for $x^2 - 5x + 6 = 0$, $-b = +5$.

- Not shifting everything to one side to get the standard form $ax^2 + bx + c = 0$ before using D or factorising.
- Cancelling a common factor containing x from both sides, which loses a root (e.g. $x(x - 2) = 3x$).
- Not rejecting a negative root for length, speed, age or time in word problems.
- Saying $D = 0$ means no roots; it means two *equal* real roots.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Quadratic Equations** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** two equal real roots	$D = 0$
2	**(b)** 9	Factorise
3	**(b)** $\frac{7}{2}$	Sum $= -b/a$
4	**(d)** $-\frac{15}{2}$	Product $= c/a$
5	**(b)** 7	Roots are 4, 11
6	**(d)** 21	$9 - 30 + k = 0$
7	**(a)** no real roots	$D = -11$
8	**(c)** 12	Factorise
9	**(b)** $\frac{19}{3}$	Sum $= -b/a$
10	**(c)** 2	Product $= c/a$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Put $x = 5$: $25 - 55 + k = 0$ (1); $k = 30$ (1); other root $= 11 - 5 = 6$ (1).	1+1+1
12	$D = 0$: $(8)^2 - 4k(2) = 0$ (1); $k = 8$ (1).	1+1
13	$D = (-7)^2 - 4(2)(-15) = 169$ (1); $x = \dfrac{7 \pm 13}{4}$ (1); roots $-\frac{3}{2}, 5$ (1).	1+1+1
14	Put $x = 3$: $9 - 30 + k = 0$ (1); $k = 21$ (1); other root $= 10 - 3 = 7$ (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Multiply by $x(x+2)$: $2x + 2 = \frac{8}{15}x(x+2)$ (1); form the quadratic with integer coefficients (1); positive root $x = 3$ (1).	1+1+1
16	$D = 0$ (1); two equal real roots (1); roots $\frac{3}{2}$ (1).	1+1+1
17	$D = (-11)^2 - 4(2)(5) = 81$ (1); $x = \dfrac{11 \pm 9}{4}$ (1); roots $x = 5, \frac{1}{2}$ (1).	1+1+1
	OR: Multiply by $x(x+1)$: $2x + 1 = \frac{9}{20}x(x+1)$ (1); form the quadratic with integer coefficients (1); positive root $x = 4$ (1).	1+1+1
18	$D = -16$ (1); no real roots (1); no real roots exist (1).	1+1+1
	OR: $D = (-8)^2 - 4(1)(-9) = 100$ (1); $x = \dfrac{8 \pm 10}{2}$ (1); roots $x = 9, -1$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Speed x: $\dfrac{420}{x} - \dfrac{420}{x+5} = 2$ (1); $2100 = 2x(x+5)$ so $x^2 + 5x - 1050 = 0$ (2); $(x-30)(x+35) = 0$ (1); speed 30 km/h (1).	1+2+1+1
	OR: Boat speed x: $\dfrac{24}{x-2} + \dfrac{24}{x+2} = 5$ (1); $24(2x) = 5(x^2 - 4)$ (1); roots of the resulting quadratic: $x = 10$ or negative (1); speed 10 km/h (1).	1+1+1+1
20	Speed x: $\dfrac{240}{x} - \dfrac{240}{x+8} = 1$ (1); $1920 = x(x+8)$ so $x^2 + 8x - 1920 = 0$ (2); $(x-40)(x+48) = 0$ (1); speed 40 km/h (1).	1+2+1+1
	OR: Boat speed x: $\dfrac{24}{x-3} + \dfrac{24}{x+3} = 6$ (1); $24(2x) = 6(x^2 - 9)$ (1); roots of the resulting quadratic: $x = 9$ or negative (1); speed 9 km/h (1).	1+1+1+1

Mistakes to avoid:

- Writing the formula as $\dfrac{-b \pm \sqrt{b^2 + 4ac}}{2a}$ or dividing only part of the numerator by $2a$.
- Forgetting the sign of b: for $x^2 - 5x + 6 = 0$, $-b = +5$.

- Not shifting everything to one side to get the standard form $ax^2 + bx + c = 0$ before using D or factorising.
- Cancelling a common factor containing x from both sides, which loses a root (e.g. $x(x - 2) = 3x$).
- Not rejecting a negative root for length, speed, age or time in word problems.
- Saying $D = 0$ means no roots; it means two *equal* real roots.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Quadratic Equations** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 8	Roots are -6, 2
2	**(c)** 42	$36 - 78 + k = 0$
3	**(c)** two distinct real roots	$D = 9$
4	**(a)** 7	Factorise
5	**(a)** $-\frac{5}{4}$	Sum $= -b/a$
6	**(c)** $-\frac{3}{2}$	Product $= c/a$
7	**(b)** 7	Roots are 1, 8
8	**(c)** 35	$49 - 84 + k = 0$
9	**(a)** no real roots	$D = -32$
10	**(c)** 13	Factorise

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$D = 0$: $(12)^2 - 4k(3) = 0$ (1); $k = 12$ (1).	1+1
12	$D = (-19)^2 - 4(3)(6) = 289$ (1); $x = \dfrac{19 \pm 17}{6}$ (1); roots $\frac{1}{3}, 6$ (1).	1+1+1
13	Put $x = 6$: $36 - 78 + k = 0$ (1); $k = 42$ (1); other root $= 13 - 6 = 7$ (1).	1+1+1
14	$D = 0$: $(10)^2 - 4k(5) = 0$ (1); $k = 5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Multiply by $x(x+2)$: $2x + 2 = \frac{5}{12}x(x+2)$ (1); form the quadratic with integer coefficients (1); positive root $x = 4$ (1).	1+1+1
16	$D = 16$ (1); two distinct real roots (1); roots $-\frac{1}{5}, -1$ (1).	1+1+1
17	$D = (4)^2 - 4(1)(-12) = 64$ (1); $x = \dfrac{-4 \pm 8}{2}$ (1); roots $x = 2, -6$ (1).	1+1+1
	OR: Multiply by $x(x+1)$: $2x + 1 = \frac{11}{30}x(x+1)$ (1); form the quadratic with integer coefficients (1); positive root $x = 5$ (1).	1+1+1
18	$D = -23$ (1); no real roots (1); no real roots exist (1).	1+1+1
	OR: $D = (-14)^2 - 4(3)(8) = 100$ (1); $x = \dfrac{14 \pm 10}{6}$ (1); roots $x = 4, \frac{2}{3}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Base x, altitude $x - 7$: $x^2 + (x-7)^2 = 169$ (1); $2x^2 - 14x - 120 = 0$, i.e. $x^2 - 7x - 60 = 0$ (2); $(x-12)(x+5) = 0$ (1); base 12 cm, altitude 5 cm (1).	1+2+1+1
	OR: Ages x and $36 - x$: $(x-4)(32-x) = 147$ (1); simplifying $x^2 - 36x + 275 = 0$ (2); $(x-25)(x-11) = 0$ (1); ages 25 and 11 years (1).	1+2+1+1
20	Base x, altitude $x - 17$: $x^2 + (x-17)^2 = 625$ (1); $2x^2 - 34x - 336 = 0$, i.e. $x^2 - 17x - 168 = 0$ (2); $(x-24)(x+7) = 0$ (1); base 24 cm, altitude 7 cm (1).	1+2+1+1
	OR: Ages x and $46 - x$: $(x-8)(38-x) = 144$ (1); simplifying $x^2 - 46x + 448 = 0$ (2); $(x-32)(x-14) = 0$ (1); ages 32 and 14 years (1).	1+2+1+1

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- Not shifting everything to one side to get the standard form $ax^2 + bx + c = 0$ before using D or factorising.
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